Yunsoo Adrienne Yoon

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Mechanical Engineering graduate student with experience in robotic system development and full product lifecycle engineering, including design, assembly, and experimental testing of hardware.

EDUCATION

Northwestern University | Evanston, IL — Sep 2025 – *Expected* Dec 2026

M.S. in Mechanical Engineering, GPA: 3.67/4.0

Relevant Courses: AI & ML in Robotics, Robotic Manipulation, Microcontroller System Design*

Cornell University | Ithaca, NY — Sep 2021 – May 2025

B.S. in Mechanical and Aerospace Engineering, GPA: 3.79/4.0, *Magna Cum Laude*

Relevant Courses: Fast Robots, Foundation of Robotics, Advanced Mechatronics, System Dynamics

PROFESSIONAL EXPERIENCE

HAND ERC, Northwestern University, *Graduate Researcher* **|** Evanston, IL — Oct 2025 – Present

- Developing a flexible, high-density tactile sensing skin for dexterous robotic hands utilizing custom capacitive sensor arrays; optimizing spatial resolution, full surface coverage, and structural integration.
- Designing and evaluating sensor architectures and material stacks for compliant robotic manipulation applications; Characterizing sensor performance for repeatability, sensitivity, hysteresis, and signal stability.
- Collaborating on a multi-author research paper establishing a benchmarking framework for robotic manipulation, defining quantitative performance metrics for dexterity.

HAPPI Lab, Cornell University, *Undergraduate Researcher* **|** Ithaca, NY — Jun 2024 – May 2025

- Engineered, prototyped, and validated a self-powered wearable system, developing bench-level test stands and procedures to evaluate performance, efficiency, and failure modes.
- Optimized an axial-flux magnetic generator using a Halbach array, making data-driven design decisions based on experimental performance results, improving power output by a factor of 300+.
- Created and maintained engineering documentation including test procedures, experimental results, and design iteration logs to support verification, traceability, and reproducibility.

BALA Consulting Engineers, *HVAC Mechanical Intern* **|** New York, NY — Jun 2023 – Aug 2023

- Ran equipment simulations using IES Virtual and validated them through hand calculations.
- Developed detailed demolition and renovation plans in AutoCAD, streamlining construction coordination.

SELECTED PROJECTS

Mini-teleoperation Robot, Northwestern University | Evanston, IL — May 2026 – Present

- Developing a wireless teleoperation robot arm platform using Micro: bit v2 microcontrollers and an SO-ARM100, implementing embedded sensing, wireless communication, and actuator control for remote robotic manipulation.
- Integrating hardware and software subsystems for responsive real-time operation.

Cornell Electric Vehicles, Cornell University, *Drivetrain & Manufacturing Lead* **|** Ithaca, NY — Sep 2021 – May 2025

- Led a major electromechanical system redesign, transitioning from a mechanical differential to a dual shaft motor architecture, improving torque transmission efficiency from approximately 26% to 86%.
- Designed and prototyped drivetrain components and assembly fixtures using CAD, machining, and additive manufacturing, incorporating GD&T, tolerance stack-ups, and assembly sequencing to improve manufacturability.
- Led and contributed to 12 structured design reviews (4 as lead), documenting design decisions, manufacturing considerations, and action items to support design maturity and production readiness.

Fast Robots: High-Speed Autonomous Navigation, Cornell University | Ithaca, NY — Jan 2025 – May 2025

- Integrated sensing, PID control, and communication subsystems for an autonomous mobile robot.
- Verified overall system performance through structured testing, debugging, and documentation.

AWARDS

Presidential Science Scholarship — 2020 - 2025

- $200,000 scholarship granted to 20 Korean students in STEM majors by the President of South Korea for outstanding academic excellence, leadership, and potential to contribute to advance scientific innovation.

Cornell University Engineering Learning Initiatives (ELI) Undergraduate Researcher Award — 2024

- $5,400 grant received for undergraduate research for demonstrating technical merit and real-world practicality.

SKILLS

Robotics & Mechatronics: Sensor fusion, sensing & actuation, embedded hardware testing, PCB design

CAD & Design: Autodesk Inventor, Fusion 360, AutoCAD, GD&T, rapid prototyping, tolerance stack-up

Manufacturing: Manual mill & lathe, manufacturing drawings, CAM, CNC, 3D printing, rapid prototyping, soldering

Programming & Software: C/C++, Python, MATLAB, Arduino IDE, ROS, Linux, Excel, ANSYS